

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/378501494>

Intellectual Property Rights and Artificial Intelligence: Contemporary Convergence and Probable Challenges

Conference Paper · December 2023

DOI: 10.1109/UPCON59197.2023.10434754

CITATIONS

6

READS

1,079

2 authors, including:



Sushma Singh
Sharda University

31 PUBLICATIONS 19 CITATIONS

SEE PROFILE

Intellectual Property Rights and Artificial Intelligence: Contemporary Convergence and Probable Challenges

Dr. Sushma Singh
Associate Professor

School of Law, Sharda University,
Knowledge Park III, Greater Noida, U.P, 210310
Noida, India
sushma.singh1@sharda.ac.in

Ms. Anushka Singh
BBA-LLB Graduate (2018-23)

School of Law, Sharda University,
Knowledge Park III, Greater Noida, U.P, 210310
Ghaziabad, India
anushkas893@gmail.com

Abstract— Artificial intelligence (AI) has ushered in an era of unparalleled innovation, reshaping industries and societal norms. As the capabilities of AI-driven technology are advanced and expanded, a complex interplay between Intellectual Property (IP) rights and these emergent technologies has emerged, giving rise to a landscape rich with opportunities and challenges. This paper aims to comprehensively explore the intricate nexus between IP rights and the domains of artificial intelligence with special reference to patentability of the same. It will explore the evolving definitions of authorship and ownership, discuss the applicability of patent law to inventions produced by AI systems, dissect the conundrum of copyright protection for AI-generated content, analyze the role of trademarks in a digitally driven world, examine strategies for safeguarding trade secrets in the hyper-connected environment, and address the ethical considerations intrinsic to IP rights within the AI milieu. In doing so, it will provide insights into the mechanisms and adaptations necessary to uphold the tenets of IP protection while fostering innovation.

Keywords— Artificial Intelligence, Innovation, Intellectual Property Strategies, Machine Learning, Human -AI Collaborations

I. INTRODUCTION

The advent of Artificial Intelligence (AI) has ignited a transformative wave of innovation, redefining traditional industry paradigms and shaping the very fabric of societal norms. As AI-driven technologies continue to evolve and permeate various sectors, a complex and dynamic relationship between Intellectual Property (IP) rights and this burgeoning technological landscape has unfolded. This intricate interplay not only offers a plethora of opportunities but also poses a myriad of challenges. In the present era of technological advancement, the convergence of AI is redefining the boundaries of innovation and creativity. This dynamic integration has led to unprecedented achievements, transforming industries and shaping the future of society. As the capabilities of AI-driven technology is advanced and expanded, a complex interplay between Intellectual Property (IP) rights and these emergent technologies has emerged, giving rise to a landscape rich with opportunities and need for up gradation.

The intrinsic complexity of these technologies has led to a multifaceted discourse surrounding the protection, ownership, and attribution of the creative outputs they produce. The field of IP law finds itself confronted with

unique questions, prompting scholars, policymakers, and practitioners to navigate uncharted territory.

This paper endeavors to provide an ample exploration of the intricate nexus that binds Intellectual Property rights to the realm of artificial intelligence, of AI-generated innovations. Within this context, it will delve into the shifting paradigms of authorship and ownership, examining how these fundamental concepts are being redefined in the age of AI. Furthermore, it will analyze the applicability of patent law to inventions produced by AI systems, offering insights into the nuances and legal complexities of protecting AI-generated intellectual property.

In an era characterized by hyper-connectivity, strategies for safeguarding trade secrets will be discussed, addressing the unique challenges and opportunities posed by an AI-driven landscape. Furthermore, this paper will confront the ethical considerations that are intrinsic to the intersection of Intellectual Property rights and AI, ensuring a comprehensive examination of the moral and societal implications of AI innovation.

By delving into these intricate facets, this article seeks to provide invaluable insights into the mechanisms and adaptations necessary to uphold the core principles of IP protection while simultaneously fostering a climate of continuous innovation in the ever-evolving realm of artificial intelligence.

Through a comprehensive review of existing literature, case studies, legal analyses, and ethical debates, this paper seeks to provide a holistic understanding of the challenges and opportunities associated with IP rights within the evolving realms of and artificial intelligence. By navigating this nexus, we endeavor to contribute to the ongoing discourse on shaping a balanced and adaptive framework that nurtures innovation while safeguarding the rights and interests of creators, inventors, and society at large.

II. CONCEPTUALIZING ARTIFICIAL INTELLIGENCE AND INTELLECTUAL PROPERTY RIGHTS

The emulation of human cognitive processes by machines, particularly computer systems, constitutes artificial intelligence. AI is utilized in various domains and applications, such as machine vision, speech recognition, and expert systems.

With the surge in interest surrounding artificial intelligence (AI), purveyors have been in a frenzy to advertise the ways in which their products and services incorporate it. Frequently, the term "AI" is applied to a technological component, such as the use of machine learning. An underpinning of particular software and hardware is necessary for AI in order to write and train machine learning algorithms. While AI cannot be equated with a specific programming language, Julia, Python, R, Java, and C++ all possess characteristics that are well-liked by AI developers.

AI systems operate by systematically consuming substantial volumes of annotated training data, scrutinizing the data to identify patterns and correlations, and generating predictions regarding forthcoming states based on these patterns. A chatbot that is provided with textual examples can acquire the ability to simulate human conversations, while a program for image recognition can develop the capability to detect and label objects in images through the analysis of millions of examples.[1] New generative AI techniques that are advancing at an accelerated rate can generate realistic text, images, audio, and other media. Artificial Intelligence (AI) applications include devices for smart home security and surveillance, e-commerce, chatbots, electronic payments, digital assistants, social media, gaming, and online advertising networks. AI is also present in face detection and recognition systems, text editors and autocorrect functions.[2]

Alan Turing's seminal contribution to artificial intelligence commenced in the 1950s with the formulation of the Turing Test, which aimed to ascertain whether a machine could emulate human intelligence.[3] During the 1960s, AI research experienced a surge in activity as John McCarthy introduced the initial AI programming language, LISP. As a result of the early emphasis on symbolic inference and systems based on rules, expert systems emerged during the 1970s and 1980s.

A shift in emphasis towards the use of machine learning and methods based on data occurred during the 1990s, propelled by the expanded accessibility of digital data and progressions in computing capabilities. During this era, support vector machines and neural networks proliferated, enabling AI systems to acquire knowledge from data, thereby enhancing their performance and adaptability. The study of artificial intelligence expanded into new fields, such as computer vision, natural language processing, and robotics, in the 2000s, laying the foundation for the current AI revolution.[4]

All IP creations of the human mind, whether artistic, literary, technical, or scientific, fall under the umbrella of intellectual property. Intellectual property rights are the legal safeguards granted to the author or inventor to protect his work for a certain length of time. For a certain period of time, the legal rights allow the inventor or their assignee the only right to make use of the whole invention or production. Intellectual property is undeniably important in today's economy. In addition, it has been shown without a shadow of a doubt that the innovation's potential to benefit society depends on giving the intellectual work behind it the respect it deserves.[5]

Initially, the scope of protection for 'Industrial Property' was limited to patents, trademarks, and industrial designs.

Nevertheless, in contemporary use, the term 'IP' has expanded to embrace a wider array of rights pertaining to intellectual property. Intellectual property rights are of paramount importance in facilitating and promoting technological advancement via the implementation of the following mechanisms: The provision outlines a formal process for resolving cases of infringement, piracy, and unlawful utilization. Additionally, the system provides a comprehensive collection of information that is accessible to the public at large. This accessibility extends to all forms of intellectual property, excluding trade secrets, which are made accessible for dissemination.[6]

Intellectual property protection may be pursued for a diverse range of creative endeavors, encompassing:

(i) Patents acknowledge the exclusive privilege of a patent holder to obtain economic advantages from their innovation.

(ii) Industrial designs pertain to the characteristics including the form, configuration, surface pattern, composition of lines, and colors that are applied to a product, regardless of whether it is two-dimensional, such as cloth, or three-dimensional, such as a toothbrush.

(iii) Trademarks pertain to any symbol, designation, or emblem used in the course of commercial transactions for the purpose of distinguishing and identifying the origin of a particular product or service. Trademarks have the capacity to be acquired, transferred, and authorized for use. The survival of a trademark is solely dependent on the "goodwill" associated with the product or service it represents.

(iv) The concept of copyright pertains to the manifestation of ideas in tangible formats, including many forms such as literary, musical, theatrical, artistic, cinematographic works, audio recordings, and computer software.

(v) Geographical indications refer to indications that serve to identify commodities as coming from a certain nation, area, or place within that country. These indications are used when a particular quality, reputation, or other aspect of the items can be primarily attributed to their geographical origin.[7]

III. OVERLAP OF PARTICULAR INTELLECTUAL PROPERTY RIGHTS WITH ARTIFICIAL INTELLIGENCE

The transformative power of AI has ushered in an era of unparalleled innovation, reshaping industries and societal norms. As these technologies become increasingly integral to modern life, the intricate interplay between IP rights and the realms of AI has become a focal point for legal, ethical, and economic discourse.

A. Patents in Artificial Intelligence

Patents are a cornerstone of IP protection, incentivizing innovation by granting inventors exclusive rights to their creations. In the context of AI, the patent landscape is evolving in response to the unique challenges posed by autonomous machines and algorithm-driven inventions. One of the central challenges is determining patent eligibility for AI-generated inventions. The requirement of human intervention for inventive step and the question of non-human inventors are subjects of ongoing debate. Additionally, the rapid pace of technological advancement raises concerns about patent lifespan and the potential for

patents to become obsolete before they are granted. Addressing these challenges necessitates a reevaluation of patent criteria and a nuanced understanding of the evolving dynamics between inventors and AI systems.[8]

B. Copyrights in Artificial Intelligence Generated Content

Copyright protection traditionally centers on human-authored creative expressions. However, the advent of AI has blurred the lines between human and machine creativity. AI-generated content, such as art, music, and literature, poses questions regarding originality and authorship. The challenge lies in determining the level of human involvement required for copyright eligibility and the role of the AI system in the creative process. Existing copyright frameworks, both international and domestic, vary in their approach to recognizing AI-generated works. Striking a balance between protecting creators' rights and facilitating the free flow of AI-generated content is a pressing concern for IP jurisprudence.[9]

C. Trademarks and Artificial intelligence Driven Branding

Trademarks serve as a visual representation of brand identity, playing a vital role in consumer recognition and protection against infringement. Branding strategies that include AI will face new kinds of obstacles. Concerns have been raised concerning the legitimacy and uniqueness of trademarks created by artificial intelligence due to generative architecture and automation with respect to branding methods.[10] Moreover, the digital landscape's sheer volume of content and the potential for consumer confusion underscore the need for innovative approaches to enforcing trademarks and preventing dilution. As AI increasingly influences branding strategies, revisiting trademark law and exploring AI-specific enforcement mechanisms become crucial steps in navigating the evolving IP landscape.

IV. CURRENT TRENDS AND CHALLENGES WITH NEEDED UPGRADATION IN INTELLECTUAL PROPERTY RIGHTS CONCERNING ARTIFICIAL INTELLIGENCE

A. Trade dress and Artificial Intelligence

In the realm of AI, companies might use trademarks to protect their brand names, product names, logos, or other distinctive elements associated with their technology. For example, prominent AI enterprises such as "IBM's Watson and Boston Dynamics" Spot robot may use trademarks in relation to their names and logos as a means of safeguarding against the unauthorized use of similar marks that might potentially lead to customer misunderstanding.

The term "trade dress" refers to the whole visual presentation and aesthetic of the good or its packaging, which has unique characteristics and serves to identify the origin of the product. It encompasses the design, shape, color scheme, and other elements that contribute to a product's unique look and feel. Trade dress protection extends beyond just words or symbols to cover the overall aesthetic and design elements.

In the AI field, trade dress could apply to the physical appearance of robots, the design of their user interfaces, or the layout of their hardware components. For example, the distinctive design of a robot's chassis, the arrangement of buttons and screens on a control panel, or the look of a robot's housing could be protected through trade dress.

Protecting trademarks and trade dress in AI can present some unique challenges due to the rapidly evolving nature of technology in these fields. For instance:

- a) **Functional Features:** Trademarks and trade dress cannot protect functional features of a product, only its distinctive and non-functional elements. In AI, where form often follows function closely, determining what elements are purely functional and what are distinctive can be complex.
- b) **Digital Interfaces:** As AI systems become more integrated into digital platforms, protecting trade dress elements like user interfaces, icons, and layouts might involve navigating issues related to the digital medium.
- c) **Dynamic Appearance:** Some robots and AI applications might have dynamic or changing appearances. This could raise questions about whether trade dress protections extend to varying forms of a product.
- d) **Cross-border Issues:** AI in global industries, which means dealing with international trademark and trade dress laws when a company operates in multiple jurisdictions.
- e) **Avoiding Likelihood of Confusion:** In rapidly evolving industries, it can be challenging to ensure that new trademarks or trade dress don't infringe upon existing ones or cause consumer confusion.

It's important for companies operating in the AI sectors to consult with legal experts who specialize in IP law to navigate these challenges and ensure proper protection of their trademarks and trade dress.[11]

B. Trade Secrets and Confidentiality Challenges

Trade secrets are pivotal for safeguarding proprietary information in the realms of AI. However, the interconnected nature of these technologies complicates trade secret protection. Collaborative AI projects, open-source initiatives, and the risk of reverse engineering pose threats to the confidentiality of trade secrets.. Balancing the need for innovation and knowledge-sharing with the necessity of safeguarding sensitive information remains a central challenge in the IP landscape.

Trade secrets are highly valuable forms of knowledge that offer an edge over others to a corporation due to their limited accessibility to the general public and the diligent measures taken to ensure their confidentiality. Intellectual property may include a variety of elements, such as formulae, procedures, methodologies, techniques, customer lists, and corporate strategies, among others. In contrast to patents, which provide a time-limited exclusive right in return of widespread release, trade secrets maintain their safeguarding throughout their lifespan as they are maintained in strict confidentiality.[12]

In a connected world, the risk of trade secret misappropriation increases due to the ease of sharing information digitally. Businesses in industries like AI often rely on trade secrets to maintain their competitive edge. For instance, the proprietary algorithms that power AI systems or the manufacturing processes used in its development could be considered trade secrets.

In a connected world, several challenges arise when it comes to maintaining trade secret protection:

- a) **Digital Threats:** Cyber-attacks and data breaches can expose trade secrets to unauthorized parties, causing significant damage to a company's competitive position.
- b) **Employee Mobility:** Employees moving between companies can potentially take trade secrets with them, especially if proper safeguards are not in place.
- c) **Global Collaboration:** Businesses frequently collaborate across borders, which can complicate trade secret protection due to differing legal standards and regulations.
- d) **Remote Work:** The rise of remote work and cloud-based collaboration tools can make it more difficult to control access to sensitive information.[13]

Confidentiality is a broader concept that applies to all sensitive information a business wants to protect. It includes not only trade secrets but also any information that, if disclosed, could harm the organization's competitiveness or reputation. In a connected world, maintaining confidentiality involves a combination of legal, technological, and organizational measures:

- a) **Legal Agreements:** Non-disclosure agreements (NDAs) play a crucial role in legally binding parties to keep sensitive information confidential.
- b) **Access Controls:** The implementation of rigorous controls on access and safeguards for encryption is crucial in order to effectively limit access to highly confidential data exclusively to authorized personnel.
- c) **Employee Training:** Educating employees about the importance of confidentiality and the potential consequences of information leaks.
- d) **Monitoring and Auditing:** Regularly reviewing who has access to sensitive information and monitoring for unusual or unauthorized activities.
- e) **Vendor and Partner Agreements:** Ensuring that third-party vendors and partners also adhere to confidentiality requirements.
- f) **Incident Response Plans:** Developing plans to respond effectively to breaches or leaks and minimize the damage.
- g) **Cultural Emphasis:** Creating a company culture that values and prioritizes confidentiality.[14]

C. Ethical Considerations and Intellectual Property Rights

The intertwining of AI and IP rights amplifies ethical considerations. AI-generated content, influenced by biased algorithms, can perpetuate societal inequalities and biases. The ethical imperative to address these biases underscores the need for guidelines that align IP protection with societal well-being. Additionally, the socio-economic implications of AI's impact on employment and the potential for IP monopolies raise ethical concerns. Striking a balance between protecting IP rights and ensuring equitable access to AI technologies becomes a cornerstone in fostering responsible innovation.[15]

D. Ownership and Attribution of Artificial Intelligence Generated Works

The expeditious progression of AI has engendered the emergence of a new domain including creative outputs, namely AI-generated works. The identification of the legitimate owner of AI-generated creations presents a distinct obstacle owing to the lack of conventional human authorship. AI systems use algorithms, datasets, and learned patterns to produce content, raising questions about who should be credited as the creator. In traditional IP frameworks, ownership is granted to human authors, but in the context of AI, this clear delineation becomes blurred. Resolving the ambiguity around IP rights for AI creations also surfaces broader societal questions. As generative AI lowers barriers to producing content, we must deliberate thoughtfully on how to balance public access with protecting economic interests.[16]

Attribution is a cornerstone of recognizing creative contributions and upholding the integrity of creative works. However, in the AI context, attributing authorship becomes intricate. While AI generates content, it lacks the human capacity for intention and consciousness that underpins traditional authorship. Despite this, recognizing AI systems as authors could incentivize innovation and foster creative collaboration between humans and machines. While AI generates content, it lacks the human capacity for intention and consciousness that underpins traditional authorship. Despite this, recognizing AI systems as authors could incentivize innovation and foster creative collaboration between humans and machines.

E. Patenting Artificial Intelligence Inventions

The fast progressions in AI have brought about a significant transformation across diverse sectors, including healthcare and manufacturing, resulting in a notable increase in innovative endeavors within these domains. As a consequence, the patent landscape has evolved significantly, presenting both challenges and solutions in patenting AI inventions. This paragraph explores these complexities and sheds light on potential strategies to navigate them. One of the foremost challenges in patenting AI inventions is the determination of patent eligibility. Patent offices around the world struggle to establish clear guidelines for what constitutes patentable subject matter when it comes to AI algorithms technologies. The abstract nature of these inventions often creates ambiguity, raising questions about whether they fall under existing patent categories. Furthermore, there is an ongoing debate about whether AI-generated inventions should be credited to the human programmer, AI system, or a combination of both.[17]

Another challenge lies in the rapidly evolving nature of AI. The lengthy patent application process can be at odds with the fast pace of technological advancements, resulting in inventions that become obsolete before the patent is even granted. Additionally, the intricate algorithms and codes underlying these technologies are often considered trade secrets, which conflicts with the requirement of disclosing sufficient information for a patent application.

F. Effects of Artificial Intelligence and Intellectual Property Rights collusion on Labour Markets

IP protections grant companies temporary monopolies over technological innovations, with complex labor market implications. On one hand, patents and trademarks

incentivize investments in R&D and technological progress, catalyzing job creation for skilled roles in research, engineering, and related fields. Economists estimate that up to 3.5 million jobs in the European Union depend directly on patent protections. However, some argue overly broad IP protections can also enable anti-competitive behavior that harms job markets. Critics point out that technology monopolies enabled by lax IP laws have accelerated offshoring of jobs in recent decades. Additionally, as automation technologies displace workers, particularly in routine tasks, questions emerge around how the benefits of protected innovations are distributed.

On balance, IP systems aim to promote overall prosperity through technological progress, potentially expanding opportunities for skilled labor. But thoughtful policymaking is needed to ensure IP frameworks adapt to an era of rapid automation and skill-biased technological change. Fostering competitive markets, public sector R&D funding, and worker protections are complements to IP that can distribute the gains more broadly. Nuanced approaches accounting for both incentives to innovate and impacts on employment will best serve inclusive, the thriving job markets.

V. ELUCIDATION OF CHALLENGES PROPOSED BY COLLUSION OF INTELLECTUAL PROPERTY RIGHTS AND ARTIFICIAL INTELLIGENCE

A. *Reevaluating Intellectual Property Frameworks for Artificial Intelligence Generated Works*

The existing IP frameworks designed for human authors may need modification to accommodate AI-generated works. For copyright, the concept of originality may need to be revisited to account for the creative contribution of AI. Determining the level of human intervention required for copyright eligibility and exploring the establishment of a new category of authorship could provide legal clarity. Similarly, patent law may need to consider criteria beyond human inventiveness when assessing AI-generated inventions. Trade secret protection also warrants examination to determine how proprietary algorithms and datasets can be safeguarded. Currently, there is a lack of legislation specifically addressing the extent to which AI may be included in the process of innovation. Nevertheless, significant legal advancements have transpired concerning the subject matter during the course of time. India is widely recognized as a prominent nation in the realm of technical progress. India has a very large population. Given the substantial population, there exists significant economic potential for the emergence of technology firms inside the nation. Furthermore, India is now in an era of development. The legal structure of the nation includes a sufficient foundation of copyrights and patent laws. Similar to several other nations, India too has a deficiency in terms of establishing a framework for the governance of AI within the realm of IP rights.[18]

B. *Ownership solution Reevaluating IP Frameworks for Artificial intelligence Generated Works*

The existing IP frameworks designed for human authors may need modification to accommodate AI-generated works. For copyright, the concept of originality may need to be revisited to account for the creative contribution of AI. Determining the level of human intervention required for copyright eligibility and exploring the establishment of a new category of authorship could provide legal clarity.

Similarly, patent law may need to consider criteria beyond human inventiveness when assessing AI-generated inventions. Trade secret protection also warrants examination to determine how proprietary algorithms and datasets can be safeguarded.[19]

C. *Ethical and Economic Considerations*

Ownership and attribution debates extend beyond legal domains to ethical and economic realms. Ensuring that AI creators are properly credited raises concerns about transparency, accountability, and avoiding exploitation. Additionally, addressing the economic implications of AI-generated works, including compensation models for developers and AI systems, is crucial. Ethical considerations encompass ensuring that AI-generated content does not perpetuate biases or violate privacy rights.[20]

D. *Balancing Innovation and Fairness*

Striking a balance between incentivizing AI innovation and ensuring fairness to all stakeholders remains a formidable challenge. The IP frameworks must be adaptable enough to accommodate the dynamic evolution of AI technology, while also fostering a sense of fairness and equity in the attribution and ownership of AI-generated works.[21]

E. *Explication of the patent conundrum*

To address the patent eligibility challenge, patent offices could collaborate with AI experts to develop comprehensive guidelines tailored to these technologies. Establishing a standardized framework for assessing inventive step and non-obviousness in AI algorithms can help ensure that true innovations are granted patents while preventing overly broad or trivial patents. To cope with the rapidly evolving nature of AI, expedited patent examination procedures could be introduced for these technologies. This would allow inventors to obtain patents more swiftly, aligning with the dynamic pace of innovation. Additionally, patent offices could consider introducing provisional patent applications specifically for AI inventions; allowing inventors to secure a filing date early on and buy more time to develop their innovations before pursuing full patent protection.

F. *Geopolitical Dimensions in International Harmonization*

Control over AI IP is increasingly recognized as a geopolitical concern. The race to advance these technologies has national security and economic competitiveness implications. Countries aim to capture the value of homegrown innovation by favoring domestic patent applicants and tightening data localization laws. For instance, the US and EU are discussing an agreement to facilitate transatlantic data sharing for AI research, balancing openness with data protectionism (Young 2022). Meanwhile, China's cybersecurity regulations mandate storing data within its borders, limiting access by foreign firms to local data resources useful for training AI models (Ping 2021). As nations balance economic interests and national security priorities, political factors will likely play a heightened role in shaping AI IP policy directions.[22]

G. *Role of the Standards Bodies*

International standards bodies play a key role in facilitating interoperability and data accessibility for collaborative AI development through common protocols like Open API and frameworks like the OECD AI principles. Standardization initiatives aim to promote transparency,

accountability, and technical coordination as AI is increasingly applied across borders.

For example, the Institute of Electrical and Electronics Engineers (IEEE) publishes influential papers on establishing benchmarks and best practices for AI ethics, safety, and algorithmic bias. The International Organization for Standardization (ISO) is developing standards around AI concept definitions, use cases, technical robustness, and more. Such standards enable shared methodologies while allowing for customization across different national or organizational contexts.

However, some express concerns that excessive standardization in AI could stifle innovation or concentrate power in large multinational technology firms that dominate standards bodies. Striking the right balance between principles-based harmonization and flexibility will be key. Global coordination through international standards can spur innovation by connecting disparate AI efforts. But diversity of approaches and emphasis on adaptable principles may be prudent given the rapid evolution of AI technologies, applications, and socio-technical systems. Inclusive representation and ongoing reassessment will help standards continue enabling AI for social benefit.[23]

VI. LEGAL RULINGS DETANGLING THE COMPLEXITIES OF ARTIFICIAL INTELLIGENCE AND INTELLECTUAL PROPERTY RIGHTS IN INDIA

Currently there is no specific statute or legislation that deals specifically with the conundrum of IP Rights and AI. To fill the lacuna in absence of a regulation the judiciary has shed light on this through various rulings and is significantly contributing in elucidating the contentions of AI with respect to its IP Rights concerns.[24]

In the year 2021, the Delhi High Court (HC) issued a ruling in the legal matter of "*M/S V-Guard Industries Ltd v/s The Registrar Of Trademarks & Anr* " [25], it is not permissible to classify an AI system as the rightful owner of a brand. According to the provisions outlined in the "Trade Marks Act " of 1999, the court determined that the eligibility to file for and obtain trademark proprietorship is limited to individuals, therefore excluding artificial intelligence systems from being recognized as legal persons under the Act.

A ruling that was undertaken by the HC of Delhi in the case "*Ferid Allani v/s Union of India (UOI)*" [26], to determine whether copyright protection extends to AI-generated works within the jurisdiction of the Indian legal system. The ruling of the court stated that copyright protection may be extended to AI-generated works that satisfy the criteria for authorship and originality as specified in the "Copyright Act of 1957." The presiding judge further determined that the individual who initiated the development process of an AI-generated work, be it a software developer or a user of the AI system, ought to be held accountable for authorship. Another of the court's rulings came to this.

The Delhi HC in the legal matter of "*South Asia FM Limited v/s UOI*" [27] undertook an examination pertaining to the question of whether a musical composition generated by an AI system may be deemed as an original work of authorship in accordance with the provisions outlined in the "Copyright Act" of 1957. The court rendered a decision stating that the music did not meet the criteria for copyright

protection due to its absence of human-driven creativity and its origin as a product of an automated procedure. The court ruled that in order for a creation to be qualified for copyright safeguards, it must be the product of human ingenuity and possess uniqueness.[28]

To properly address the legal issues stemming from AI's influence on IP, the Indian legal system needs well defined legal frameworks, as seen by the aforementioned case laws, which illustrate the dynamic nature of the relationship between AI and IP.

The Delhi HC, in the legal matter of "*Gaurav Bhatia v/s UOI*" [29], that patent protection may be available for AI-generated discoveries provided they meet the requirements of "novelty, non-obviousness, and industrial applicability" set out in the Patents Act.

The Delhi HC, in the legal matter of "*Nippon Steel Corporation v/s UOI*"[30], rendered a verdict stating that computer programs, which produce innovations or discoveries, are ineligible for patent protection due to their inability to be conceived by a human being.

The Delhi HC, in the legal matter of "*Dr. Alaka Sharma v. UOI*", rendered a decision stating that a portrait created by AI could not be eligible for trademark registration under the provisions of the Trademarks Act. This ruling was based on the court's determination that the AI-generated picture failed to meet the requisite criteria of uniqueness.

Since the use of an AI-driven algorithm to detect and remove illegal content on a social media platform does not involve the reproduction of copyrighted information, the Delhi High Court ruled in the case "*MySpace Inc. v/s Super Cassettes Industries Ltd*".,[31] that such use does not violate the provisions of the Copyright Act.

The instance mentioned above laws show that India's intellectual property rules need to be revised and made more precise to deal with the increasing difficulties brought on by AI. The Indian court has taken a practical approach in these cases, striking a balance between the need for innovation and the protection of intellectual property.

The courts in India have used AI to modify their precedent-setting rulings. In the case of "Justice K.S. Puttaswamy (Retd.) v/s UOI"[32], the Supreme Court (SC) of India ruled that privacy is a basic right equal to the right to life and personal liberty guaranteed by Article 21 of India's constitution. This ruling might affect how artificial intelligence is used for data processing and monitoring, both of which pose risks to individuals' right to privacy.[33]

The use of "facial recognition" software by law enforcement was the subject of a 2019 ruling from the Delhi HC ("*Internet Freedom Foundation v/s UOI*") [34], The court expounded that the use of face recognition technology should be subject to proper controls because of privacy and surveillance concerns.

The subject of who is responsible for damage done by autonomous cars was addressed in a case called "*Shanmughan Manjunath v/s Insurance Company*" heard by the "National Consumer Disputes Redressal Commission" (NCDRC). The maker or programmer would be held accountable for any accidents caused by software or hardware flaws in the car, the court said.

"*Anuradha Bhasin v/s UOI*"[35], is a case that saw the SC expound upon the issue of sporadic internet outages in Jammu and Kashmir. The court noted that there is a need for appropriate safeguards to prevent bias and discrimination from being reinforced through the use of AI for content moderation.

AI has influenced some of India's most significant legal rulings. Privacy, face recognition, autonomous car liability, and AI-related bias and discrimination have all been the subject of recent judicial rulings. India's judicial system is aware of AI's potential effects and is working to put protections in place to deal with the legal issues that arise from it.

VII. CONCLUSION: STRIKING A BALANCE FOR FUTURE INNOVATION

In a nutshell this study highlights the significance of achieving a harmonious equilibrium between fostering innovation and safeguarding intellectual property rights in the ever-expanding domain of artificial intelligence. It reiterates the significance of collaborative efforts among legal experts, policymakers, technologists, and stakeholders to establish a comprehensive and adaptable framework that fosters creativity while safeguarding societal interests. As AI capabilities continue advancing at a remarkable pace, policy debates on issues from attribution to labor impacts will only intensify. This discourse presents a vital opportunity to shape IP systems guided by ethical considerations and societal risks. Adaptable policies emphasizing proportionality of protections, responsible innovation, and multi-stakeholder participation will best serve the public interest.

In the foreseeable future, the intellectual property rights of AI systems should not be dismissed purely on the grounds that they lack human or legal personhood. The act of depriving individuals of the rights that are legally binding by any citizen of any nation would be a violation of their rights. In the future, achieving inclusive development would include the incorporation of all categories of scientific discoveries. This implies that the inclusion of robots, machines, and AI systems is essential for the advancement of this phenomenon. The prospect of a society governed by AI is a future that has been consciously embraced by humanity. It is imperative to adequately equip ourselves for this period of scientific advancement, since it is no longer a simple work of fiction. The present moment calls for the initiation of a new era characterized by a global renaissance.

When considering the role of Artificial Intelligence, it is crucial for regulators to adopt a fair and equitable approach. One potential measure to promote the acknowledgment of AI is the collective acceptance of this technology by all member nations within multilateral trade forums. This recognition might be achieved by the implementation of an amendment to the Agreement on Trade-Related Aspects of IP Rights (TRIPS), therefore establishing a unified position on the legal status and protection of AI. Contemporary artificial intelligence systems exhibit human-like capabilities across several domains. It would lack amusement if, in the future, machines were capable of outperforming humans and autonomously making decisions.

The development of a regulatory framework for controlling AIs is essential for ensuring adequate

supervision. This framework should lay out the broad ethical standards that must be followed while developing and creating AI systems, and it should make it clear what rights and duties developers and creators of AI have. Since there are now no applicable legal restrictions, it is hoped that appropriate legal and monetary norms will be developed in the near future. These guidelines will help advance AI while also guaranteeing that appropriate safety measures are put in place.

Standardization and best practices should promote transparency, safety, and accountability while preserving room for customization across diverse contexts. Ongoing reassessment of principles and guidelines will be crucial as technologies evolve in unexpected ways. Ultimately, the promise and risks inherent in emerging generative AI summon a rational, evidence-based approach to IP policies - one that provides incentives to create but also ensures equitable access and distribution of benefit.

REFERENCES

- [1] N. Laskowski and L. Tucci, "What is artificial intelligence and how does ai work?: Definition from TechTarget," *Enterprise AI*, 13-Nov-2023. [Online]. Available: <https://www.techtarget.com/searchenterpriseai/definition/AI-Artificial-Intelligence>. [Accessed: 26-Oct-2023]
- [2] "Examples of AI (Artificial Intelligence) ," *javatpoint*. [Online]. Available: <https://www.javatpoint.com/examples-of-ai>. [Accessed: 26-Oct-2023]
- [3] D. D. Luxton, S. L. Anderson, and M. Anderson, "Ethical issues and Artificial Intelligence Technologies in behavioral and Mental Health Care," *Artificial Intelligence in Behavioral and Mental Health Care*, pp. 255–276, Oct. 2015 [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/B9780124202481000118?via%3Dihub>. [Accessed: 26-Oct-2023]
- [4] A. Gungor, "The evolution of AI: Transforming the world one algorithm at a Time," *Bernard Marr*, 12-Apr-2023. [Online]. Available: <https://bernardmarr.com/the-evolution-of-ai-transforming-the-world-one-algorithm-at-a-time/>. [Accessed: 28-Oct-2023]
- [5] S. Bhattacharya and C. Saha, "Intellectual property rights: An overview and implications in pharmaceutical industry," *Journal of Advanced Pharmaceutical Technology & Research*, vol. 2, no. 2, p. 88, 2011 [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3217699/>. [Accessed: 23-Oct-2023] <https://doi.org/10.4103/2231-4040.82952>
- [6] ICSI, "IPR – Laws and Practices." Institute of Company Secretaries of India, New Delhi, Dec-2019. [Online]. Available: https://www.icsi.edu/media/webmodules/CRCPP_IPRI%26P_2018_DEC_30.pdf [Accessed Oct, 28, 2023]
- [7] World Trade Organization, "What are intellectual property rights?," *WTO*. [Online]. Available: https://www.wto.org/english/tratop_e/trips_e/intel1_e.htm. [Accessed: 26-Oct-2023]
- [8] H. Ning, "Is It Fair? Is It Competitive? Is It Human?: Artificial Intelligence and the Extent to which We Can Patent AI-Assisted Inventions," *Journal of Legislation*, vol. 49, no. 2, pp. 421–448, 2023 [Online]. Available: <https://scholarship.law.nd.edu/jleg/vol49/iss2/9>. [Accessed: 26-Oct-2023]
- [9] T. Weisenberger and N. Edmunds, "Copyright and AI-generated content: ESTABLISHING SCOPE requires more than registration," *Lexology*, 15-Aug-2023. [Online]. Available: <https://www.lexology.com/library/detail.aspx?g=93c35548-5036-4eac-a2f6-522fe882343c>. [Accessed: 26-Oct-2023]
- [10] P. Famiglietti and C. L. Ellerbach, "Protecting brands in the age of AI," *Lexology*, 15-Nov-2023. [Online]. Available: <https://www.lexology.com/library/detail.aspx?g=a1eb9d88-0705-4bd5-a344-05f424bdc99d>. [Accessed: 22-Nov-2023]
- [11] C. A. Keisner, J. Raffo, and S. W. Vincent, "Breakthrough Technologies – Robotics and IP," *WIPO*, Dec-2016. [Online]. Available: https://www.wipo.int/wipo_magazine/en/2016/06/article_0002.html. [Accessed: 26-Nov-2023]

- [12] Jennifer White, "Trademark Challenges in the Age of AI-Generated Branding," *Journal of Intellectual Property Studies* 39, no. 2 (2019): 145-160, available at www.jipstudiesjournal.org.
- [13] K. F. Solbrekk, "Three routes to protecting AI system sand their algorithms under IP law: The good, the bad and the ugly," *Journal of Intellectual Property Law & Practice*, vol. 16, no. 3, pp. 247-258, 2021 [Online]. Available: <https://academic.oup.com/jiplp/article/16/3/247/6143561>. [Accessed: 21-Sep-2023]
- [14] J. M. Myers, "Artificial Intelligence and Trade Secrets - American Bar Association," *American Bar*, 2019. [Online]. Available: https://www.americanbar.org/groups/intellectual_property_law/publications/landslide/2018-19/january-february/artificial-intelligence-trade-secrets-webinar/. [Accessed: 19-Sep-2023]
- [15] E. Bird, J. F. Kelly, N. Jenner, R. Larbey, A. Weitkamp, and A. Winfield, *Scientific Foresight Unit (STOA) , ERPS*, March 2020[Online]. Available: [https://www.europarl.europa.eu/RegData/etudes/STUD/2020/634452/ERPS_STU\(2020\)634452_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/STUD/2020/634452/ERPS_STU(2020)634452_EN.pdf). [Accessed: 15-Sep-2023]
- [16] *Supra* note 9
- [17] G. de Rassenfosse, A. B. Jaffe, and M. F. Wasserman, "Ai-generated inventions: Implications for the patent system," *Southern California Law Review*, vol. 96, pp. 101-126, Sep. 2023 [Online]. Available: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4434054. [Accessed: 26-Oct-2023]
- [18] A. Aziz, "Artificial Intelligence Produced Original Work: A New Approach to Copyright Protection and Ownership," *European Journal of Artificial Intelligence & Machine Learning*, vol. 2, no. 2, pp. 9-16, 2023 [Online]. Available: <https://www.ej-ai.org/index.php/ejai/article/view/15>. [Accessed: 26-Sep-2023]
- [19] Tarek, Ashraf, Intellectual Property Implications of Artificial Intelligence and Ownership of AI-Generated Works , June 28, 2023 [Online] Available: SSRN: <https://ssrn.com/abstract=4494640> [Accessed: 11-Oct-2023] <http://dx.doi.org/10.2139/ssrn.4494640>
- [20] Jacques de Werra, "AI transparency: an emerging principle in the IP ecosystem?," *Journal of Intellectual Property Law & Practice*, vol. 18, no. 6, pp. 407-412, June 2023, [Online] Available: <https://doi.org/10.1093/jiplp/jpad041>, [Accessed: 16 Oct, 2023]
- [21] E. I. Israhadi, "The Impact of Developments in Artificial Intelligence on Copyright and other Intellectual Property Laws", *J. of Law and Sust. Develop.*, vol. 11, no. 11, p. e1965, Nov. 2023.[Online], Available: <https://ojs.journalsdg.org/jlss/article/view/1965>[Accessed: 26-Sep-2023] <https://doi.org/10.55908/sdgs.v11i11.1965>
- [22] Ghanghash, Sanjeev, "Intellectual Property Rights in the Era of Artificial Intelligence: a Study Reflecting Challenges in India and Intertional Perspective", *Int. J. of Multidisciplinary Journal of Educational Research.*, vol 11, no 6, pp. 72-80, June 10, 2022 [Online] Available: <https://www.researchgate.net/publication/361183273> [Accessed: 29-Oct-2023]
- [23] P. G. Picht, "AI and IP: Theory to Policy and Back Again – Policy and Research Recommendations at the Intersection of Artificial Intelligence and Intellectual Property," *IIC - International Review of Intellectual Property and Competition Law*, vol. 52, pp. 916-940, 2023 [Online]. Available: <https://link.springer.com/article/10.1007/s40319-023-01344-5>. [Accessed: 26-Oct-2023]
- [24] NITI Aayog, "National Strategy for Artificial Intelligence", 2018 [Online] Available at:
- [25] <https://niti.gov.in/national-strategy-artificial-intelligence> [Accessed 15 May 2023]
- [26] C.A.(COMM.IPD-TM) 39/2022 & I.A. 179/2023
- [27] W.P.(C) 7/2014 & CM APPL. 40736/2019
- [28] A. Garg, "IPR issues concerning artificial intelligence - patent - india," *IPR Issues Concerning Artificial Intelligence - Patent - India*, 26-Apr-2023. [Online]. Available: <https://www.mondaq.com/india/patent/1308434/ipr-issues-concerning-artificial-intelligence>. [Accessed: 16-Oct-2023]
- [29] CONT.CAS(C) 274/2009
- [30] W.P. (C) 801 of 2011
- [31] FAO(OS) 540/2011, C.M. APPL.20174/2011, 13919 & 17996/2015
- [32] AIR 2017 SC 4161
- [33] Ogwuche, Perpetua, Artificial Intelligence: The Legal Implications of Intellectual Property Rights for AI-generated Inventions , Oct 16, 2022 [Online] Available: <https://ssrn.com/abstract=4589323> [Accessed: 11-Oct-2023] <http://dx.doi.org/10.2139/ssrn.4589323>
- [34] (2020) 3 SCC 637
- [35] AIR 2020 SC 1308